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Highlights

- An analytical solution of axial dispersion model in a fixed bed column
- Binary and multicomponent mixture adsorption in a fixed bed column
- Transient concentration and temperature profiles in the fixed-bed column.

**Concentration and temperature profiles in a fixed bed column
based on an analytical solution of the axial dispersion model
for binary and multicomponent non-isothermal adsorption processes**

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Abstract

An analytical solution to model dynamic adsorption of multicomponent mixtures in packed beds is presented. The analytical solution describes transient concentration profiles together with gas and solid phase temperature profiles during adsorption and desorption processes. The breakthrough curves predicted by the analytical solution are in satisfactory agreement with numerical simulations and experimental data reported in the literature.

Keywords Fixed bed, Modeling, Binary adsorption, Multicomponent adsorption, Breakthrough

1. Introduction

Adsorption in a fixed bed column includes multicomponent mass and heat transfer processes combined with adsorption and desorption processes. The dynamic balance equations describe transient concentration and temperature profiles in the adsorption column. The modeling of such processes often relies on the numerical integration of the partial differential equations describing the transfer processes in fixed-bed adsorption, which in certain cases results in numerical instabilities and long calculation times. A number of analytical solutions of those systems of partial differential equations for the axial dispersion model (ADM) have been reported in the literature, but they are often limited to isothermal conditions and/or single-component systems (Ruthven, 1984; Tien, 1994). Cooney and Strusi (1972) proposed an analytical solution for isothermal binary mixture adsorption in a fixed bed column defined by a plug flow model with axial dispersion. Miura et al. (1979) developed a method for calculating the binary mixture breakthrough curves in a fixed-bed column using an analytical solution of partial differential equations describing the transfer processes. Liaw et al. (1979) obtained an analytical solution of the partial differential equations for isothermal adsorption based on the plug flow model taking into account the transfer processes in the adsorbent particle. Rasmuson and Neretnieks (1980, 1985) proposed an analytical solution of an axial dispersion model in a fixed bed column with linear sorption in the particles. Seidel-Morgenstern (1991) considered a fixed-bed adsorber with a linear adsorption isotherm and used Laplace inversion with a Fourier series expansion to obtain an analytical solution and to determine the effect of boundary conditions on the axial dispersion model. Liao and Shiau (2000) applied the separation of variables method to an axial dispersion model of a fixed-bed adsorber with a linear adsorption isotherm and derived an analytical solution.

A typical model of adsorption in the fixed-bed column includes (i) description of transfer processes within an adsorbent pellet; (ii) balance equation for the adsorbate in the fluid phase; and (iii) adsorption equilibrium equation coupling concentration of a component in fluid and solid phase. Weber and Chakravorti (1974) provided a comparison of the analytical solution derived by Thomas

(1944) and Hiester and Vermuellen (1952), the homogeneous solid phase diffusion model of Rosen (1952), the linear driving force method of Glueckauf (1955) and the pore phase diffusion model (Edeskuty and Amundson, 1952; Kasten et al., 1952). They reviewed the various possible solutions for the general case of a fixed bed with a homogeneous solid or pore diffusion model for the pellets. For the case of constant pattern behavior with uniform velocity for the established temperature and concentration fronts in the fixed-bed column, Ruthven et al. (1975) obtained the analytical solution of the non-isothermal plug flow model with Langmuir adsorption isotherm.

The purpose of this study is to obtain an analytical solution for the case of binary and multicomponent adsorption in a fixed bed column taking into account the coupled heat and mass transfer processes based on the axial dispersion model.

2. Axial dispersion model of adsorption

The axial dispersion model (ADM) is a well-known model, based on an idealization of the flow patterns in a fixed-bed column. The axial dispersion model of transfer processes developed in the current work is based on the following assumptions: (i) ideal gas flow in the column; (ii) Langmuir type adsorption isotherm; (iii) a Linear Driving Force (LDF) model is used to describe mass transfer in the solid phase; (iv) there is thermal equilibrium at the solid-gas interface; (v) mass and heat transfer in the radial direction are negligible; (vi) there is a symmetrical concentration front profile relative to the stoichiometric time. A theoretical framework for the concentration front during adsorption in a fixed bed column and the effect of the equilibrium isotherm has been given by De Vault (1943) and Tondeur (1987).

The axial dispersion model balance equations are defined as shown in Table 1. The developed axial dispersion model is based on dynamic balance equations taking into account the adsorbed components concentration, temperature and component concentration in the gas phase. The source terms in Table 1 take into account the interface transfer processes in fixed bed column.

According to the LDF adsorption model, the driving force of the adsorption process is proportional to the difference between the local (q) and equilibrium concentrations (q_{eq}) in the adsorbed phase. The heat of adsorption (ΔH_{ads}) is taken into account in the heat balance equations. The initial condition is set in accord with the operating conditions of the adsorption column:

$$\text{Adsorption: } t=0, F^{(m)} = F_{ini}^{(m)}, q^{(m)} = q_{ini}^{(m)}, T^G = T_{ini}^G, T^S = T_{ini}^S.$$

Table 1

3. Analytical solution of the axial dispersion model

The main variables of the axial dispersion model in Table 1 are

- Component molar flux $F^{(m)} = F^{(m)}(t, z)$;
- Adsorbate concentration $q^{(m)} = q^{(m)}(t, z)$;
- Gas-phase temperature $T^G = T^G(t, z)$;
- Solid phase temperature $T^S = T^S(t, z)$.

The integration of this system of partial differential equations is complicated by the fact that differential equations are non-linear with the coupled variables reflecting the coupled heat and mass transfer processes. The driving idea in this work is to simplify the differential equations using the relationship between the variables

$$\frac{\partial y^{(m)}}{\partial t} = \frac{\partial y^{(m)}}{\partial q^{(m)}} \frac{\partial q^{(m)}}{\partial t}, \quad \frac{\partial T^G}{\partial t} = \frac{\partial T^G}{\partial T^S} \frac{\partial T^S}{\partial t}$$

As shown in the Appendix, the modification of the conservation equations allows us to derive an analytical expression for the driving force of adsorption (ψ). Using the obtained analytical expression (1) we are able to integrate the mass and heat transfer balance equations in Table 1. The resulting component concentration and temperature profiles are calculated using numerical integration:

$$\begin{cases} \tau > 0, \psi_{i,j}^{(m)} = \psi_{ref}^{(m)} 0.5 \exp(-k_f^{(m)} \tau) \\ \tau < 0, \psi_{i,j}^{(m)} = \psi_{ref}^{(m)} (0.5 + 0.5(1 - \exp(-k_f^{(m)} \tau))) \end{cases} \quad (1)$$

$$F_{i,j}^{(m)} = F_{i,0}^{(m)} - (1 - \varepsilon_b) \rho_b k_f^{(m)} \sum_{k=0}^j \gamma_q^{(m)} \psi_{i,j}^{(m)} \Delta z \quad (2)$$

$$Q_{i,j}^G = Q_{i,0}^G + \varepsilon_b \alpha_v^G \sum_{k=0}^j (\Delta T_{i,j}) \Delta z \quad (3)$$

$$\Delta T_{i,j} = \frac{(1 - \varepsilon_b) \rho_b \sum_k (-\Delta H_{ads}^{(k)}) k_f^{(k)} \psi_{i,j}^{(k)}}{\alpha_v^G} \quad (4)$$

$$y_{i,j}^{(m)} = \exp(Pe_G z_j / L) \left(y_{i,0}^{(m)} - \frac{Pe_G}{\rho_{mol}^G \varepsilon_b u_G L} \sum_{k=0}^j F_{i,k}^{(m)} \exp(-Pe_G z_k / L) \Delta z \right) \quad (5)$$

$$T_{i,j}^G = \exp(Pe_T z_j / L) \left(T_{i,0}^G - \frac{Pe_T}{C_p \rho_{mol}^G \varepsilon_b u_G L} \sum_{k=0}^j Q_{i,k}^G \exp(-Pe_T z_k / L) \Delta z \right) \quad (6)$$

$$q_{i,j}^{(m)} = q_{eq,i,j-1}^{(m)} - \psi_{i,j}^{(m)} \quad (7)$$

$$T_{i,j}^S = T_{i,j}^G + \Delta T_{i,j} \quad (8)$$

$$\psi_{ref}^{(m)} = q_{eq,i,j-1}^{(m)} - q_{ini,i,j-1}^{(m)} \quad (9)$$

where t_i - time, $t_i = i \Delta t$; τ - the corrected time, $\tau = (t_i - z_j / u_f^{(m)}) - t_s^{(m)}$; $t_s^{(m)}$ - stoichiometric breakthrough time; z_j - coordinate, $z_j = j \Delta z$. The location of mesh point is designed by (i) in time-domain and (j) in z-direction.

The assumption of a symmetric profile (1) is valid for the low concentration of the adsorbed component in the gas phase with $\gamma = \text{const}$. For $\gamma = \gamma(t)$ at high concentrations of the adsorbed component in the gas phase, we can use an approximation of the analytical solution in the form of a time series as shown in Appendix. The coefficient (γ_q) takes into account the influence of the shape of an adsorption isotherm on the driving force of adsorption. The definition of the coefficient (γ_q) is given in the Appendix. The boundary conditions for model equations (1)-(8) are specified at the inlet section of the column

$$F_{i,0}^{(m)} = F_t y_{in}^{(m)}, \quad Q_{i,0}^G = F_t C_p T_{in}^G, \quad y_{i,0}^{(m)} = y_{in}^{(m)}, \quad T_{i,0}^G = T_{in}^G$$

For a binary gas mixture adsorption, the empirical Sips equation defines equilibrium adsorbed concentration as a function of temperature, pressure, and composition

$$q_{eq,i,j}^{(m)} = \frac{q_{\max}^{(m)} (b_m P y_{i,j}^{(m)})^{t_m}}{1 + \sum_{k=1}^2 (b_k P y_{i,j}^{(k)})^{t_k}} \quad (10)$$

The obtained equation for driving force of adsorption (1) describes sorption front propagation in the fixed bed column. The component concentration and temperature profiles in the column are calculated taking into account the sorption front propagation in the fixed bed layer by the following algorithm:

- (i) For the local time $t_i = i\Delta t$, the position of the sorption front is determined by equation (1);
- (ii) For the local time $t_i = i\Delta t$, the component flux profile is calculated using equation (2) from the input section ($z=0$) to the output section ($z=L$);
- (iii) For the local time $t_i = i\Delta t$, the component concentration profile is calculated using equation (5) from the input section ($z=0$) to the output section ($z=L$);
- (iv) For the local time $t_i = i\Delta t$, the heat flux profile is calculated using equation (3) from the input section ($z=0$) to the output section ($z=L$);
- (v) For the local time $t_i = i\Delta t$, the temperature profile is calculated using equation (6) from the input section ($z=0$) to the output section ($z=L$);
- (vi) Repeating steps (i)-(v) for time step $t_{i+1} = (i+1)\Delta t$ until the adsorption process completion.

The adsorption equilibrium is described using a multicomponent Sips isotherm equation (10), but in principle, any kind of equilibrium model could be applied.

4. Simulation Results and Discussion

Experimental breakthrough curves reflect the dynamics of the transfer process in fixed bed columns. The shape of the breakthrough curve is a result of multicomponent mass transfer processes coupled with adsorption and desorption processes in the column. The literature provides a large amount of

experimental data for fixed bed columns equipped with a different adsorbent (particles) under different operating conditions. We used experimental data reported by Möller et al. (2017) with the most comprehensive input data for axial dispersion model (ADM) parameters for a gas mixture consisting of CO₂/CH₄/He and using the Shirasagi MSC CT Carbon Molecular Sieve adsorbent. The chosen system corresponds to the case of Langmuir type binary adsorption from the gas phase by a solid adsorbent. The necessary input data for calculating breakthrough curves are listed in Table 2.

Table 2

The capacity of a fixed-bed adsorber depends on both the adsorption equilibrium and adsorption kinetics. The experimental breakthrough curves provide the most complete information including the kinetics of adsorption, desorption, and mass and heat transfer processes. Figure 1 shows the comparison of experimental breakthrough curves with those predicted by our analytical solution of the axial dispersion model (ADM). The solid line corresponds to the prediction by the analytical solution using the input parameters of Table 2. Points correspond to the experimental breakthrough curves reported by Möller et al. (2017). The shape of the breakthrough curve is associated with equation (1) defining the shape of the sorption front moving with the front velocity (u_f) in the fixed bed. The analytical solution is valuable for understanding the interactions of adsorption equilibrium and mass transfer processes in establishing the shape of the breakthrough curves. The component concentration in the gas phase (y) is calculated using equation (5) with component flux (2), taking into account $Pe_G = 10$ and plug flow condition ($Pe_G = \infty$). The Peclet number (Pe) reflects the influence of the flow patterns on the concentration profile in the column. Although a quite good match was obtained when assuming plug flow ($D_{ax} = 0$, red curve), a better fit was obtained using a Peclet number for mass transfer equal to 10 (green curve), reflecting the true flow patterns in the fixed bed. The model equations in Table 1 correspond to a one dimensional model with partial differential equations. For comparison, the results of a full numerical simulation of the system of

partial differential equations have been added (blue dashed curve) in Figure 1. The numerical solution of the axial dispersion model (NADM) has been obtained by implementing the model equations (Table 1) in COMSOL software.

The analytical solution (eqns. 1-8) and the numerical solution of the model coincide nicely in the first part of the breakthrough curve, but the numerical solution shows a more tailed profile. The comparison indicates that the change in Peclet criterion from ideal mixing ($Pe_G = \infty$) to $Pe_G = 10$ results in changing the shape and shifting the temperature response curve.

Fig. 1.

Aguilera and Gutiérrez Ortiz (2016) have simulated H_2S adsorption from biogas in a fixed-bed column using Comsol Multiphysics software. They have also performed a sensitivity analysis of the model with respect to the operational parameters. They have found that a change in the axial dispersion coefficient leads to changing the shape and shifting the response curve for the concentration. The general trend in the response curves in Fig.1 is consistent with the simulation results reported by Aguilera and Gutiérrez Ortiz (2016). Figure 2 shows the variation of gas phase composition, temperature and adsorption rate with time at the outlet of the column. The adsorbate concentration is calculated from equation (7). The rate of adsorption (Fig.2, middle) is defined as follows:

$$r_{ads}^{(m)} = k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) , \quad m = CO_2, CH_4 \quad (11)$$

In this system, CH_4 diffuses very slowly into the pores of the carbon molecular sieve adsorbent (see Table 2) and is nearly not adsorbed in dynamic conditions, explaining the immediate breakthrough in the experimental data and simulations. The model very well predicts both composition profiles. The rate of adsorption of CH_4 remains at a nearly constant and low value while that of CO_2 increases steeply as soon as CO_2 reached the end of the adsorption bed and starts to adsorb locally. When the pores are almost saturated with CO_2 and the mole fraction of CO_2 approaches the feed composition

after about 18 min., the adsorption rate starts to decrease. Heat is released during adsorption of CO_2 , resulting in a temporary increase in temperature of the gas and solid adsorbent in the column (Fig. 2, right). The temperature difference between gas and the solid phase is determined by the volumetric heat transfer coefficient in accord with equation (4) taking into account the heat of adsorption. The high value of the volumetric transfer coefficient (α_v^G) is associated with the high interfacial area of the fixed-bed column ensuring the close values of gas and solid phase temperatures.

Fig. 2.

The difference between equilibrium and working adsorbate concentrations is the driving force of the adsorption process (Fig.3). In accordance with the adopted computation algorithm, the calculation is performed ranging the time from $t=0$ to the specified process time by integrating the conservation equations with respect to fixed bed volume from the input section ($z=0$) to the output section ($z=L$).

Fig. 3.

The development of the component concentration profiles in the gas phase and the adsorbed concentration profiles is shown in Fig.4. The component molar flux is calculated through numerical integration of equation (2) using the analytical expression for the driving force of adsorption (1). The component concentration in the gas phase (y) is defined using equation (5) as a function of the component flux. The component concentration in the adsorbed phase is associated with the driving force of adsorption (1) by equation (7). The development of the concentration profile in the gas phase indicates the displacement of CO_2 adsorption front in fixed bed column, with slow mass transfer resulting in relatively broad concentration profiles. The concentration of CH_4 in the adsorbed phase is almost constant along the length of the column but increases steadily with time due to even much slower mass transfer in the adsorbed phase for CH_4 as compared to CO_2 .

Fig. 4.

As shown in Appendix, the driving force of heat transfer in the gas phase (4) is calculated through integrating the heat balance equation with respect to the heat flux in the gas phase (3). The obtained heat flux is integrated to calculate the temperature profile in the gas phase (6). The development of the temperature and heat flow profiles in Fig.5 and Fig.6 is indirectly associated with the displacement of the sorption front due to the release of the heat of adsorption. The heat flux at the solid interface (Fig.6) characterizes the mass and heat transfer processes coupled with adsorption and desorption in a fixed bed column. The temperature of the gas and the solid are identical in these calculations, as a result of the large value of the heat transfer coefficient (see also Fig. 6).

Fig. 5.**Fig. 6.**

Available analytical solutions in the literature are often limited by the isothermal condition and the number of components in the adsorption process. The new analytical solution is based on an integration of the conservation equations using the new analytical solution for the driving force of adsorption. The obtained analytical solution of the axial dispersion model (ADM) is suitable for binary and multicomponent non-isothermal adsorption and desorption processes in fixed bed columns. The main advantages of the derived analytical solution of the axial dispersion model are

- Flexibility in the description of multicomponent adsorption and desorption processes;
- Flexibility in defining the adsorption equilibrium;
- Flexibility in the description of the coupled transfer processes without iterations;
- Flexibility in modeling adsorption process in fixed bed column with different flow patterns;
- Description of the transient concentration and temperature profiles in a fixed bed column.

Flexibility in adsorption equilibrium is due to the fact that any equilibrium model (explicit or implicit models such as IAST) can be expressed through the equilibrium adsorbed concentration as a function of concentration, temperature and time. The proposed analytical solution is applicable for multicomponent adsorption and desorption processes. Further modification of this analytical solution is possible in future taking into account additional effects (such as pressure and temperature variations applied in PSA and TSA) on multicomponent adsorption and desorption processes in fixed bed columns.

5. Conclusions

An analytical solution of the axial dispersion model (ADM) is obtained for binary and multicomponent mixture adsorption in a fixed bed column in non-isothermal conditions. The axial dispersion model (ADM) balance equations and analytical solution describe the transient concentration and temperature profiles in the fixed-bed column. The predicted transient concentration and temperature profiles are consistent with those reported in the literature.

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Nomenclature

C_p	Heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
D_{ax}	Axial dispersion coefficient ($\text{m}^2 \text{s}$)
F	Molar flux ($\text{mol m}^{-2} \text{s}^{-1}$)
F_t	Inlet molar flux ($\text{mol m}^{-2} \text{s}^{-1}$)
ΔH_{ads}	Heat of adsorption (J mol^{-1})
k_f	LDF coefficient (s^{-1})
L	Height of fixed bed (m)
n	Number of components
P	Pressure (Pa)
Pe_G	Peclet number for mass transfer ($Pe_G = u_G L/D_{ax,G}$)
Pe_T	Peclet number for heat transfer ($Pe_T = u_G L/D_{ax,T}$)
R	Gas constant ($R=8.314 \text{ J K}^{-1} \text{mol}^{-1}$)
q	Adsorbate concentration in the adsorbed phase (mole kg^{-1})
Q	Heat flux ($\text{W m}^{-2} \text{K}^{-1}$)
T	Temperature (K)
t	Time (s)
t_s	Stoichiometric breakthrough time (s)
u_f	Sorption front velocity (m s^{-1})
u_G	Gas velocity (m s^{-1})
y	mole fraction in the gas phase
z	Coordinate (m)

Greek symbols

α_v	Volumetric heat transfer coefficient ($\text{W m}^{-3} \text{K}^{-1}$)
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$$\gamma_q \quad \text{Coefficient, } \gamma_q = \left(1 - \frac{\rho_{mol}^G \varepsilon_b}{\rho_b (1 - \varepsilon_b)} \left(\frac{\partial q}{\partial q_{eq}} \frac{\partial q_{eq}}{\partial y} \right)^{-1} \right)$$

$$\gamma_\psi \quad \text{Coefficient, } \gamma_\psi = 1 + \frac{(1 - \varepsilon_b) \rho_b}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}}{\partial y} (\gamma_q - 1)$$

ε_b The porosity of the fixed bed ($\text{m}^3 \text{ m}^{-3}$)

ρ_{mol} Molar density (mol m^{-3})

ρ Density (kg m^{-3})

τ The corrected time, $\tau = (t - z / u_f) - t_s$ (s)

ψ Driving force of adsorption (mol kg^{-1})

λ_{ax} Thermal dispersion coefficient ($\text{W m}^{-1} \text{ K}^{-1}$)

Δt Time step (s)

Δz Mesh size (m)

Sub/superscripts

ads adsorption

b fixed bed

eq equilibrium

G gas

f front

in inlet

ini initial

m component

mol molar

S solid

T temperature
i,j Space-time mesh

Abbreviations

IAST	Ideal Adsorbed Solution Theory
PSA	Pressure swing adsorption
TSA	Temperature swing adsorption
AADM	Analytical solution of the axial dispersion model
NADM	Numerical solution of the axial dispersion model
ADM	Axial dispersion model
LDF	Linear Driving Force

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Appendix A

Modification of conservation equations

The mass balance equation for the component describes the component concentration profile along the height of the column

$$\varepsilon_b \rho_{mol}^G \frac{\partial y^{(m)}}{\partial t} = \rho_{mol}^G \varepsilon_b \frac{\partial}{\partial z} \left(D_{ax} \frac{\partial y^{(m)}}{\partial z} \right) - \rho_{mol}^G \frac{\partial (u_G y^{(m)})}{\partial z} + (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t} \quad (A.1)$$

$$\frac{\partial q^{(m)}}{\partial t} = k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) \quad (A.2)$$

To analyze the mass transfer process, we write the balance equation (A.1) through the component flux

$$F^{(m)} = \rho_{mol}^G u_G y^{(m)} - \rho_{mol}^G \varepsilon_b D_{ax} \frac{\partial y^{(m)}}{\partial z} \quad (A.3)$$

Taking Eq. (A.3) into account, Eq. (A.1) is written as

$$\varepsilon_b \rho_{mol}^G \frac{\partial y^{(m)}}{\partial t} = - \frac{\partial F^{(m)}}{\partial z} + (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t} \quad (A.4)$$

We can modify the component mass balance equation (A.4) using the relation

$$\frac{\partial y^{(m)}}{\partial t} = \frac{\partial y^{(m)}}{\partial q^{(m)}} \frac{\partial q^{(m)}}{\partial t} \quad (A.5)$$

Substituting (A.5) into (A.4), we re-write the mass balance equation in the form

$$\varepsilon_b \rho_{mol}^G \frac{\partial y^{(m)}}{\partial q^{(m)}} \frac{\partial q^{(m)}}{\partial t} = - \frac{\partial F^{(m)}}{\partial z} + (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t}$$

or

$$\frac{\partial F^{(m)}}{\partial z} = (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t} \left(1 - \frac{\rho_{mol}^G \varepsilon_b}{\rho_b (1 - \varepsilon_b)} \frac{1}{\partial q^{(m)} / \partial y^{(m)}} \right) \quad (A.6)$$

To determine the partial derivative in parentheses in (A.6), we can use the relation

$$\frac{\partial q^{(m)}}{\partial y^{(m)}} = \frac{\partial q^{(m)}}{\partial q_{eq}^{(m)}} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \quad (A.7)$$

Using equation (A.2) and (A.7), the differential equation (A.6) can be written in the form

$$\frac{\partial F^{(m)}}{\partial z} = (1 - \varepsilon_b) \rho_b k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) \gamma_q^{(m)} \quad (A.8)$$

where $\gamma_q^{(m)}$ - coefficient $\gamma_q^{(m)} = \left(1 - \frac{\rho_{mol}^G \varepsilon_b}{\rho_b (1 - \varepsilon_b)} \left(\frac{\partial q^{(m)}}{\partial q_{eq}^{(m)}} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \right)^{-1} \right)$. The coefficient $\gamma_q^{(m)}$ takes into

account the influence of the shape of the adsorption equilibrium curve $(\partial q_{eq}^{(m)} / \partial y^{(m)})$ on the driving force of adsorption. The evaluation of the coefficient indicates that $\gamma_q^{(m)} \approx 1$ for the system in Table 2.

The assumption $\gamma_q^{(m)} \approx 1$ is valid for Langmuir type system adsorption in a fixed bed column under the low concentration of the adsorbed component in the gas phase. The boundary condition at the inlet section is defined for component flux as

$$u_G \rho_{mol}^G y_{in}^{(m)} = \left(u_G \rho_{mol}^G y^{(m)} - \varepsilon_b D_{ax} \rho_{mol}^G \frac{\partial y^{(m)}}{\partial z} \right) \Big|_{z=0} \quad (A.9)$$

or

$$F^{(m)}(t, 0) = F_t y_{in}^{(m)}$$

The heat balance equation describes the temperature profile in the gas phase along the column height

$$\varepsilon_b \rho_{mol}^G C_p^G \frac{\partial T^G}{\partial t} = \varepsilon_b \frac{\partial}{\partial z} \left(\lambda_{ax}^G \frac{\partial T^G}{\partial z} \right) - \rho_{mol}^G C_p^G \frac{\partial (u_G T^G)}{\partial z} + \varepsilon_b \alpha_v (T^S - T^G) \quad (A.10)$$

To analyze the transport equation, we write the balance equation (A.10) through the heat flux

$$Q^G = \rho_{mol}^G C_p^G u_G T^G - \varepsilon_b \lambda_{ax}^G \frac{\partial T^G}{\partial z} \quad (A.11)$$

The substitution of (I.11) in Eq. (I.10) gives

$$\varepsilon_b \rho_{mol}^G C_p^G \frac{\partial T^G}{\partial t} = - \frac{\partial Q^G}{\partial z} + \varepsilon_b \alpha_v (T^S - T^G) \quad (A.12)$$

Heat balance equation for the solid phase is

$$\rho^S C_p^S \frac{\partial T^S}{\partial t} = \alpha_v^G (T^G - T^S) + (1 - \varepsilon_b) \rho_b \sum_{k=1}^n (-\Delta H_{ads}^{(k)}) \frac{\partial q^{(k)}}{\partial t} \quad (A.13)$$

Under the condition of thermal equilibrium, the heat flux from the gas phase to the interface is equal to the heat flux from the interface to the solid phase. Assuming thermal equilibrium at the solid-gas interface ($\partial T_s / \partial t = 0$), we get

$$0 = \alpha_v^G (T^G - T^S) + (1 - \varepsilon_b) \rho_b \sum_{k=1}^n (-\Delta H_{ads}^{(k)}) \frac{\partial q^{(k)}}{\partial t} \quad (\text{A.14})$$

or

$$\Delta T = \frac{(1 - \varepsilon_b) \rho_b \sum_{k=1}^n (-\Delta H_{ads}^{(k)}) k_f^{(k)} (q_{eq}^{(k)} - q^{(k)})}{\alpha_v^G} \quad (\text{A.15})$$

where ΔT - temperature difference at the interface, $\Delta T = T^S - T^G$.

Using the assumption of thermal equilibrium ($\partial T_s / \partial t = 0$) and relation $\frac{\partial T^G}{\partial t} = \frac{\partial T^G}{\partial T^S} \frac{\partial T^S}{\partial t}$ for the transformation of the heat balance equation (A.12), we obtain

$$\begin{cases} \varepsilon_b \rho_{mol}^G C_p^G \frac{\partial T^G}{\partial T^S} \frac{\partial T^S}{\partial t} = -\frac{\partial Q^G}{\partial z} + \varepsilon_b \alpha_v^G \Delta T \\ \frac{\partial T^S}{\partial t} = 0 \end{cases}$$

or

$$\frac{\partial Q^G}{\partial z} = \varepsilon_b \alpha_v^G \Delta T \quad (\text{A.16})$$

The boundary condition at the inlet section is defined as

$$Q^G(t, 0) = F_t C_p^G T_{in}^G \quad (\text{A.17})$$

The component molar flux and heat profiles in the fixed bed column are obtained by integrating, respectively, Eq. (A.8) and Eq. (A.16)

$$F^{(m)}(t, z) = F_t y_{in}^{(m)} - (1 - \varepsilon_b) \rho_b k_f^{(m)} \int_0^z \gamma_q^{(m)} (q_{eq}^{(m)} - q^{(m)}(t, z)) dz \quad (\text{A.18})$$

$$Q^G(t, z) = C_p^G F_t T_{in}^G + \varepsilon_b \alpha_v^G \int_0^z \Delta T(t, z) dz \quad (\text{A.19})$$

For the axial dispersion model, the concentration and temperature profiles are obtained by integrating, respectively, Eq. (A.18) and Eq. (A.19)

$$y^{(m)}(t, z) = \exp(Pe_G z / L) \left(y_{in}^{(m)} - \frac{Pe_G}{\rho_{mol}^G \epsilon_b u_G L} \int_0^z F^{(m)}(t, z) \exp(-Pe_G z / L) dz \right) \quad (A.20)$$

$$T^G(t, z) = \exp(Pe_T z / L) \left(T_{in}^G - \frac{Pe_T}{C_p^G \rho_{mol}^G \epsilon_b u_G L} \int_0^z Q^G(t, z) \exp(-Pe_T z / L) dz \right) \quad (A.21)$$

For the plug flow model, the following relationships are valid

$$y^{(m)}(t, z) = \frac{F^{(m)}(t, z)}{\sum_{k=1}^n F^{(k)}(t, z)} \quad , \quad T^G(t, z) = \frac{Q^G(t, z)}{C_p^G \rho_{mol}^G \sum_{k=1}^n F^{(k)}(t, z)} \quad (A.20)$$

Appendix B

Driving force of adsorption

According to the modified axial dispersion model (ADM), the component mass balance equation links component molar flux (F) and adsorbent concentration (q) at the time (t)

$$\varepsilon_b \rho_{mol}^G \frac{\partial y^{(m)}}{\partial t} = -\frac{\partial F^{(m)}}{\partial z} + (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t} \quad (B.1)$$

The kinetics of adsorption is described by the LDF adsorption submodel

$$\frac{\partial q^{(m)}}{\partial t} = k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) \quad (B.2)$$

Multiplying both sides of equation (B.1) by $\frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{1}{\varepsilon_b \rho_{mol}^G}$ we obtain

$$\frac{1}{\varepsilon_b \rho_{mol}^G} \varepsilon_b \rho_{mol}^G \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial y^{(m)}}{\partial t} = -\frac{1}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial F^{(m)}}{\partial z} + \frac{1}{\varepsilon_b \rho_{mol}^G} (1 - \varepsilon_b) \rho_b \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial q^{(m)}}{\partial t} \quad (B.3)$$

Subtracting equation (B.2) from (B.3) we get

$$\frac{\varepsilon_b \rho_{mol}^G}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial y^{(m)}}{\partial t} - \frac{\partial q^{(m)}}{\partial t} = \left\{ -\frac{1}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial F^{(m)}}{\partial z} + \frac{(1 - \varepsilon_b) \rho_b}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial q^{(m)}}{\partial t} \right\} - k_f^{(m)} (q_{eq}^{(m)} - q^{(m)})$$

or

$$\frac{\partial q_{eq}^{(m)}}{\partial t} - \frac{\partial q^{(m)}}{\partial t} = \left\{ -\frac{1}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial F^{(m)}}{\partial z} + \frac{(1 - \varepsilon_b) \rho_b}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} \frac{\partial q^{(m)}}{\partial t} \right\} - k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) \quad (B.4)$$

Taking into account Eq. (B.2) and $\frac{\partial F^{(m)}}{\partial z} = \gamma_q^{(m)} (1 - \varepsilon_b) \rho_b k_f^{(m)} (q_{eq}^{(m)} - q^{(m)})$, equation (B.4) takes the form

$$\frac{\partial (q_{eq}^{(m)} - q^{(m)})}{\partial t} = -\gamma_\psi^{(m)} k_f^{(m)} (q_{eq}^{(m)} - q^{(m)}) \quad (B.5)$$

where $\gamma_\psi^{(m)}$ coefficient, $\gamma_\psi^{(m)} = 1 + \frac{(1 - \varepsilon_b) \rho_b}{\varepsilon_b \rho_{mol}^G} \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}} (\gamma_q^{(m)} - 1)$.

The evaluation shows that assumption $\gamma_q^{(m)} \approx 1$ and $\gamma_\psi^{(m)} = 1$ is valid for a Langmuir type system in Table 2 in the modeling of mass transfer processes during adsorption in a fixed bed column. Introducing variable $\psi^{(m)} = (q_{eq}^{(m)} - q^{(m)})$, we obtain

$$\frac{\partial \psi^{(m)}}{\partial t} = -k_f^{(m)} \psi^{(m)} \quad (\text{B.6})$$

or

$$\frac{d\psi^{(m)}}{\psi^{(m)}} = -k_f^{(m)} dt \quad (\text{B.7})$$

Integrating equation (B.7), we get an analytical solution of differential equation (B.2)

$$\psi^{(m)} = C \exp(-k_f^{(m)} t) \quad (\text{B.8})$$

where $\psi^{(m)}$ - the driving force of adsorption, $\psi^{(m)} = (q_{eq}^{(m)} - q^{(m)})$.

Based on the analytical solution (B.8) and assumptions about the symmetry of the concentration front profile relative to the stoichiometric time, we introduce the dimensionless driving force

$$\text{Before stoichiometric time} \quad \text{For } \tau > 0 \quad \frac{\psi^{(m)}}{\psi_{ref}^{(m)}} = 0.5 \exp(-k_f^{(m)} \tau) \quad (\text{B.9})$$

$$\text{Stoichiometric time} \quad \text{For } \tau = 0 \quad \frac{\psi^{(m)}}{\psi_{ref}^{(m)}} = 0.5 \quad (\text{B.10})$$

$$\text{After stoichiometric time} \quad \text{For } \tau < 0 \quad \frac{\psi^{(m)}}{\psi_{ref}^{(m)}} = 0.5 + 0.5(1 - \exp(-k_f^{(m)} \tau)) \quad (\text{B.11})$$

where t - time; τ - the corrected time, $\tau = (t - z/u_f^{(m)}) - t_s^{(m)}$ - stoichiometric breakthrough time; z - coordinate; $u_f^{(m)}$ - adsorption front velocity. The reference driving force ($\psi_{ref}^{(m)}$) corresponds to the maximum driving force

$$\psi_{ref}^{(m)} = (q_{eq}^{(m)} - q_{ini}^{(m)}) \quad (\text{B.12})$$

It is evident that the shape of the reference driving force $\psi_{ref}^{(m)} = \psi_{ref}^{(m)}(t)$ is following the shape of the concentration curve $y^{(m)} = y^{(m)}(t)$ due to the equilibrium concentration in the solid phase linked to

the concentration in the gas phase: $q_{eq}^{(m)} = f(y^{(m)})$. Before a stoichiometric time ($\tau > 0$) the component concentration in the gas phase is approaching zero in the mass transfer zone. After a stoichiometric time ($\tau < 0$) the component concentration in the gas phase is approaching the bulk concentration in the mass transfer zone. The local concentration in the solid phase is calculated from the driving force as follows

$$q^{(m)} = q_{eq}^{(m)} - \psi^{(m)} \quad (B.13)$$

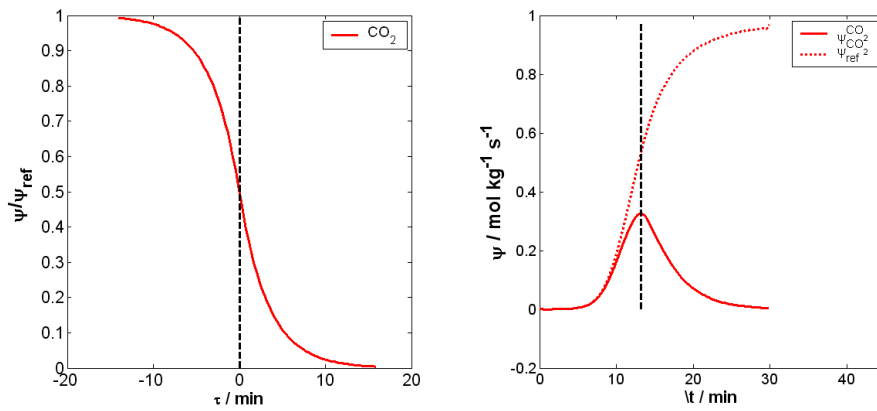


Figure B.1 Dimensionless driving force of adsorption (left) and reconstructed driving force of adsorption (right) based on an analytical solution (B.8).

Time series approximation

It should be noted that equations (B.6) - (B.11) are obtained for the case when $\gamma_\psi = const$. In general, the coefficient (γ_ψ) is a function of time. For $\gamma_\psi = \gamma_\psi(t)$, the differential equation (B.5) takes the form

$$\frac{\partial(q_{eq}^{(m)} - q^{(m)})}{\partial t} = -\gamma_\psi(t) k_f^{(m)}(q_{eq}^{(m)} - q^{(m)})$$

or

$$\frac{d\psi^{(m)}}{\psi^{(m)}} = -\gamma_{\psi}(t) k_f^{(m)} dt \quad (\text{B.14})$$

Integrating equation (B.14), we get an analytical solution of the differential equation

$$\psi^{(m)} = C \exp\left(-\int k_f^{(m)} \gamma_{\psi}(t) dt\right) \quad (\text{B.15})$$

Function (B.15) can be approximated by a time series

$$\psi^{(m)} = C \exp\left(\sum_{k=0}^n A_k \tau^{k+1}\right) \quad (\text{B.16})$$

To calculate the partial derivative $\partial q^{(m)} / \partial y^{(m)}$ in (A.6), we can use the analytical solution (B.16)

$$\frac{\partial(q_{eq}^{(m)} - q^{(m)})}{\partial y^{(m)}} = \frac{\partial(C \exp(\sum_{k=0}^n A_k \tau^{k+1}))}{\partial y^{(m)}} = 0$$

or

$$\frac{\partial q^{(m)}}{\partial y^{(m)}} = \frac{\partial q_{eq}^{(m)}}{\partial y^{(m)}}, \quad \frac{\partial q_{eq}^{(m)}}{\partial q^{(m)}} = 1 \quad (\text{B.17})$$

List of Tables

Table 1 Balance equations for axial dispersion model (ADM)

Table 2 Input data of axial dispersion model (ADM).

Table 1

Balance equations for axial dispersion model (ADM)

Balance equation	Expression
Component mass balance for gas phase	$\varepsilon_b \rho_{mol}^G \frac{\partial y^{(m)}}{\partial t} = \varepsilon_b \rho_{mol}^G \frac{\partial}{\partial z} \left(D_{ax} \frac{\partial y^{(m)}}{\partial z} \right) - \rho_{mol}^G \frac{\partial (u_G y^{(m)})}{\partial z} + (1 - \varepsilon_b) \rho_b \frac{\partial q^{(m)}}{\partial t}$
Adsorbed component	$\frac{\partial q^{(m)}}{\partial t} = k_f^{(m)} (q_{eq}^{(m)} - q^{(m)})$
Heat balance for gas phase	$\varepsilon_b \rho_{mol}^G C_p^G \frac{\partial T^G}{\partial t} = \varepsilon_b \frac{\partial}{\partial z} \left(\lambda_{ax}^G \frac{\partial T^G}{\partial z} \right) - \rho_{mol}^G C_p^G \frac{\partial (u_G T^G)}{\partial z} + \varepsilon_b \alpha_v^G (T^S - T^G)$
Heat balance for fixed bed	$\rho^S C_p^S \frac{\partial T^S}{\partial t} = -\alpha_v^G (T^S - T^G) + (1 - \varepsilon_b) \rho_b \sum_k (-\Delta H_{ads}^{(k)}) \frac{\partial q^{(k)}}{\partial t}$

Table 2

Input data of axial dispersion model (ADM).

Parameter	Value	Ref.
System	CO ₂ /CH ₄ /He/Shirasagi MSC CT	Möller et al. (2017)
CO ₂ mole fraction	0.05	Möller et al. (2017)
CH ₄ mol fraction	0.15	Möller et al. (2017)
Feed flow rate F_t (mol s ⁻¹)	1.89×10^{-3}	Möller et al. (2017)
Temperature (K)	293	Möller et al. (2017)
Pressure P (MPa)	0.5	Möller et al. (2017)
Fixed bed porosity ε_b (m ³ m ⁻³)	0.34	Möller et al. (2017)
Bulk density ρ_b (kg m ⁻³)	588.5	Möller et al. (2017)
Bed height L (mm)	200	Möller et al. (2017)
Column diameter (mm)	30	Möller et al. (2017)
CO ₂ LDF coefficient $k_f^{CO_2}$ (s ⁻¹)	1.3×10^{-2}	Möller et al. (2017)
CH ₄ LDF coefficient $k_f^{CH_4}$ (s ⁻¹)	1.2×10^{-4}	Möller et al. (2017)
He LDF coefficient k_f^{He} (s ⁻¹)	0	set
Sip equation parameter* $q_{max}^{CO_2}$ (mol kg ⁻¹)	$4.263 \exp(0.8303 (1-T/293))$	Möller et al. (2017)
Sip equation parameter* b_{CO_2} (Pa ⁻¹)	$1.013 \times 10^{-5} \exp(21.5 \times 10^3 (1/T-1/293)/R)$	Möller et al. (2017)
Sip equation parameter* t_{CO_2}	$0.7176 + 0.5077 (1-293/T)$	Möller et al. (2017)
Sip equation parameter* $q_{max}^{CH_4}$ (mol kg ⁻¹)	$3.082 \exp(1.4639 (1-T/293))$	Möller et al. (2017)
Sip equation parameter* b_{CH_4} (Pa ⁻¹)	$2.42 \times 10^{-5} \exp(9.6 \times 10^3 (1/T-1/293)/R)$	Möller et al. (2017)
Sip equation parameter* t_{CH_4}	$0.7730 + 0.6792 (1-293/T)$	Möller et al. (2017)
Pe number for mass transfer, Pe_G	10	set
Pe number for heat transfer, Pe_T	10	set
CO ₂ Heat of adsorption $\Delta H_{ads}^{CO_2}$ (J mol ⁻¹)	-21.5×10^3	Möller et al. (2017)
CH ₄ Heat of adsorption $\Delta H_{ads}^{CH_4}$ (J mol ⁻¹)	-9.6×10^3	Möller et al. (2017)
Heat transfer coefficient α_v^G (W m ⁻³ K ⁻¹)	1.68×10^5	set
Adsorbent specific heat C_p^S (J kg ⁻¹ K ⁻¹)	880	Möller et al. (2017)
Adsorbent solid phase density ρ^S (kg m ⁻³)	1.20×10^3	set

*Eq (10)

Figure captions

Fig. 1. Breakthrough curves for CO₂ concentration (left) and gas phase temperature (right) at the outlet section of the fixed bed column predicted by the analytical solution of the axial dispersion model (AADM) for CO₂/CH₄/He/Shirasagi MSC CT system as defined in Table 2. Circles represent experimental breakthrough data (Möller et al.,2017). AADM – analytical solution of axial dispersion model. NADM – numerical solution of axial dispersion model.

Fig. 2. Variation of CO₂ and CH₄ gas phase concentration (left), the rate of adsorption (middle) and temperature (right) with time at the outlet of the fixed bed column predicted by the analytical solution of the ADM for CO₂/CH₄/He/Shirasagi MSC CT system (Table 2).

Fig. 3. The variation of actual and equilibrium concentration in the adsorbed phase (left), molar flow (right) with time at the outlet of the column predicted by the analytical solution of the ADM for binary mixture adsorption. CO₂/CH₄/He/Shirasagi MSC CT system as defined in Table 2.

Fig. 4. The development of the component concentration profiles in the gas phase (left) and the adsorbed concentration (middle, right) in the fixed bed column predicted by the analytical solution of the ADM for binary mixture adsorption. CO₂/CH₄/He/Shirasagi MSC CT system as defined in Table 2.

Fig. 5. The development of the gas phase temperature profiles (left), the solid phase temperature profiles (middle) and total molar flow profiles (right) for binary mixture adsorption in the fixed bed column predicted by the analytical solution of the ADM for binary mixture adsorption. CO₂/CH₄/He/Shirasagi MSC CT system as defined in Table 2.

Fig. 6. The development of the heat flux profiles (left) and the temperature difference profiles (right) at the solid interface in the fixed bed column predicted by the analytical solution of the ADM for binary mixture adsorption. Temperature difference $\Delta T = T_s - T_G$. CO₂/CH₄/He/Shirasagi MSC CT system (Table 2).

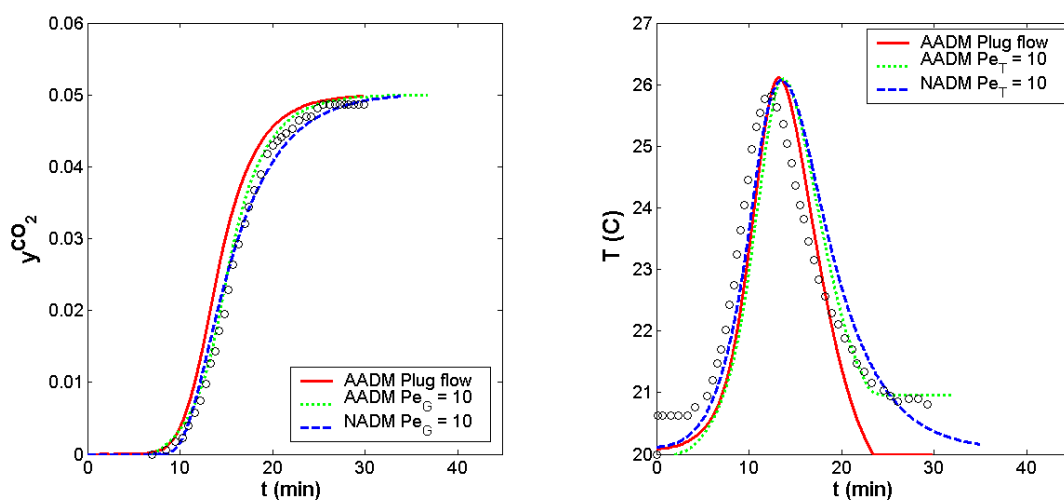


Fig. 1. Breakthrough curves for CO₂ concentration (left) and gas phase temperature (right) at the outlet section of the fixed bed column predicted by the analytical solution of the axial dispersion model (AADM) for CO₂/CH₄/He/Shirasagi MSC CT system as defined in Table 2. Circles represent experimental breakthrough data (Möller et al., 2017). AADM – analytical solution of axial dispersion model. NADM – numerical solution of axial dispersion model.

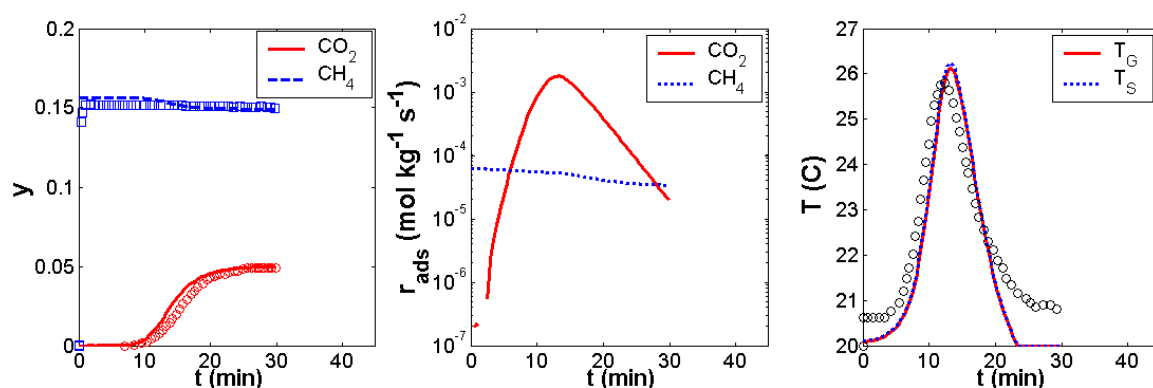


Fig. 2. Variation of CO_2 and CH_4 gas phase concentration (left), the rate of adsorption (middle) and temperature (right) with time at the outlet of the fixed bed column predicted by the analytical solution of the ADM for $\text{CO}_2/\text{CH}_4/\text{He}/\text{Shirasagi}$ MSC CT system (Table 2).

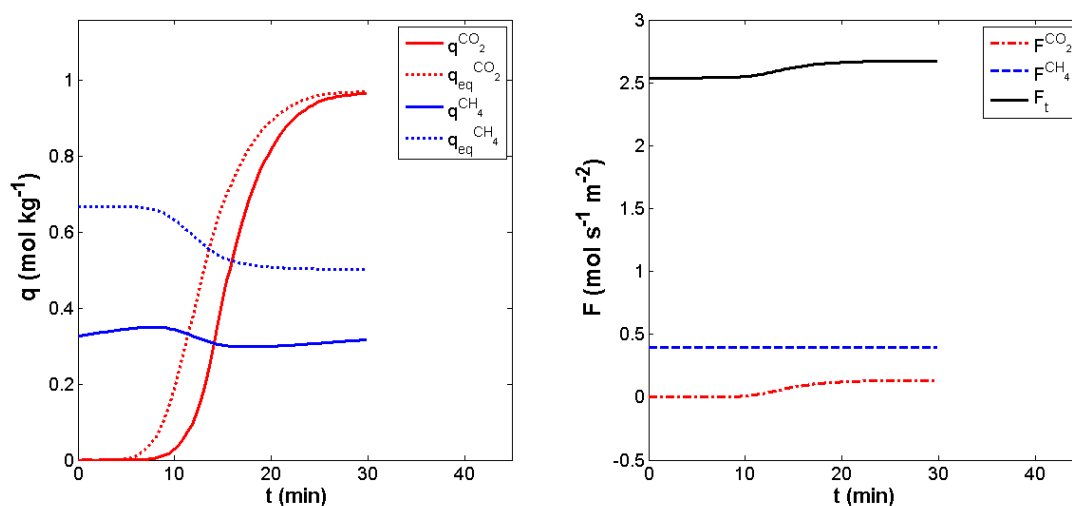


Fig. 3. The variation of actual and equilibrium concentration in the adsorbed phase (left), molar flux (right) with time at the outlet of the column predicted by the analytical solution of the ADM for binary mixture adsorption. $\text{CO}_2/\text{CH}_4/\text{He}/\text{Shirasagi}$ MSC CT system as defined in Table 2.

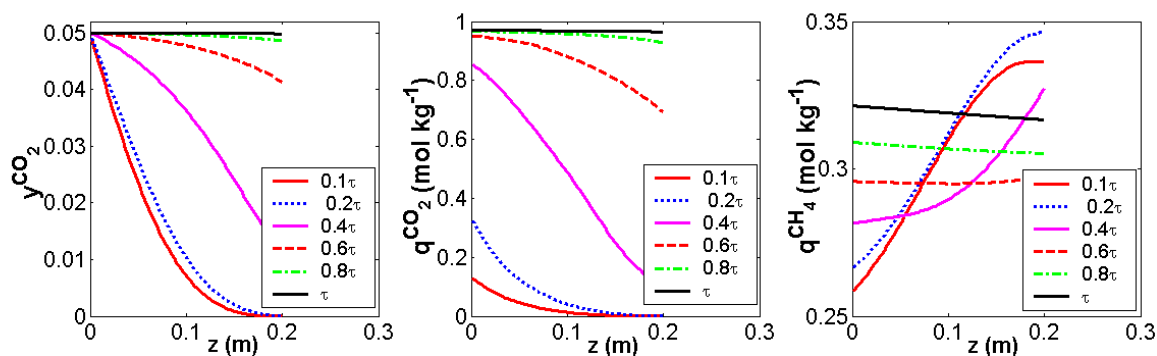


Fig. 4. The development of the component concentration profiles in the gas phase (left) and the adsorbed concentration (middle, right) in the fixed bed column predicted by the analytical solution of the ADM for binary mixture adsorption. $\text{CO}_2/\text{CH}_4/\text{He}/\text{Shirasagi MSC CT}$ system as defined in Table 2.

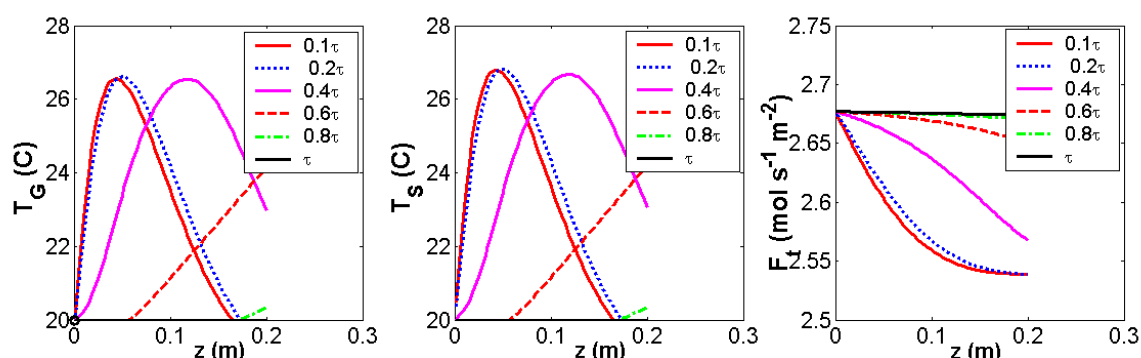


Fig. 5. The development of the gas phase temperature profiles (left), the solid phase temperature profiles (middle) and total molar flow profiles (right) for binary mixture adsorption in the fixed bed column predicted by the analytical solution of the ADM for binary mixture adsorption. $\text{CO}_2/\text{CH}_4/\text{He}/\text{Shirasagi MSC CT}$ system as defined in Table 2.

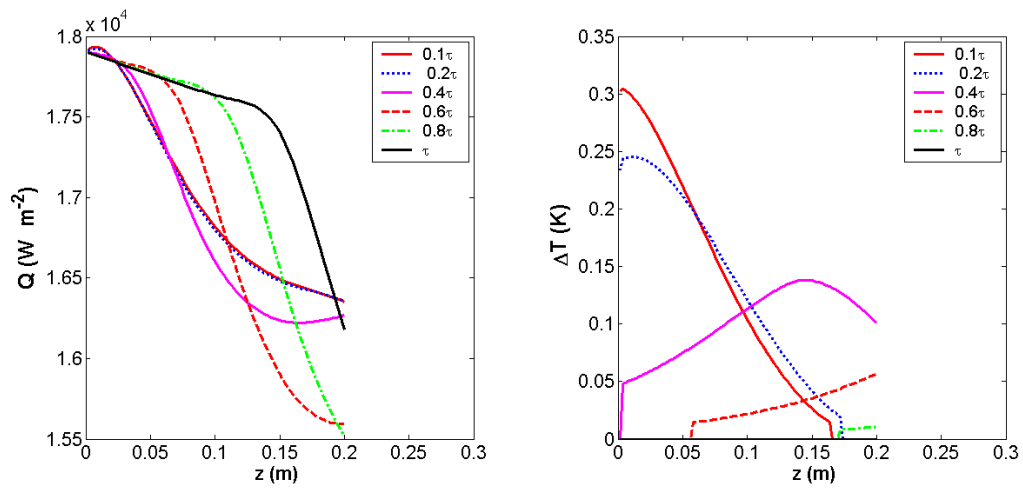


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